INCH-POUND
MIL-PRF-10304/18B
18 January 2008
SUPERSEDING
MIL-M-10304/18A(SHIPS)
9 February 1962

PERFORMANCE SPECIFICATION SHEET

METERS, MULTISCALE, DC, PANEL TYPE, RUGGEDIZED (ROUND FLANGE, 3.5 INCH), STYLE 36)

Inactive for new design after November 4, 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-10304.

REQUIREMENTS:
(Readings expressed as a percent of full-scale value.)

Dimensions and configuration: See figure 1.

Scale:
Angle: 80 degrees minimum.
Length: 2 inches minimum.

Color: The color shall be as specified in table I.

Full-scale ranges: Full-scale ranges shall be as specified in table I.

OFF-Battery marking: When applicable, the scale marking line separating “OFF” and “BATT” shall be located at the mid-scale position.

Letters and numerals: Except as specified in table I, letters and numerals shall be black.

Zero adjustment: The zero adjuster shall be rotatable through 360 degrees of angle.

Scale changing shaft: The following tests shall be performed:

Shaft rotation: Determine the shaft rotation, in degrees, required to easily read the different scales.

Torque required: Determine the torque required to rotate the shaft.

These tests shall be added and conducted in accordance with MIL-PRF-10304, as the last tests in groups I, III, and IV for qualification inspection, in group A for conformance inspection, and after the shock test in group C for qualification verification inspection.

When tested as above: The following shaft requirements shall be as specified.

Shaft rotation: The scale changing shaft (centrally located on the back of the meter) shall rotate 30 degrees for each scale position, at which position the scale shall be legible in the meter face window.

Shaft torque: The scale changing shaft shall operate with a torque of no greater than 12 ounce inches.

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Position influence: The position influence shall not exceed 3 percent of the full scale length, as a maximum change for a 60 degrees rotation from normal operating position.

Initial accuracy: The initial accuracy error shall not exceed 2 percent of full scale value.

Overshoot (Damping factor): 2.5 minimum. See MIL-PRF-10304 section 4 titled: “Overshoot”, for methods of measurement and calculation.

Response time: 2 seconds maximum.

Heat effect at + 65 degrees C:
Momentary: 4 percent of full scale value maximum.
Permanent change: 2 percent of full scale value maximum.

Temperature influence: ± 1 percent maximum.

Exposure to extreme temperatures:
Momentary: The change in indication at -55 degrees C shall not exceed 5 percent of full scale value.
Permanent: The change in indication shall not exceed 3 percent of full scale value.

Overload:
Short time: 1 percent of full scale value maximum.
Sustained:

Temporary zero shift:	1 percent of full scale maximum.
Permanent zero shift:	1 percent of full scale maximum.
Permanent change in indication:	2 percent of full scale value.

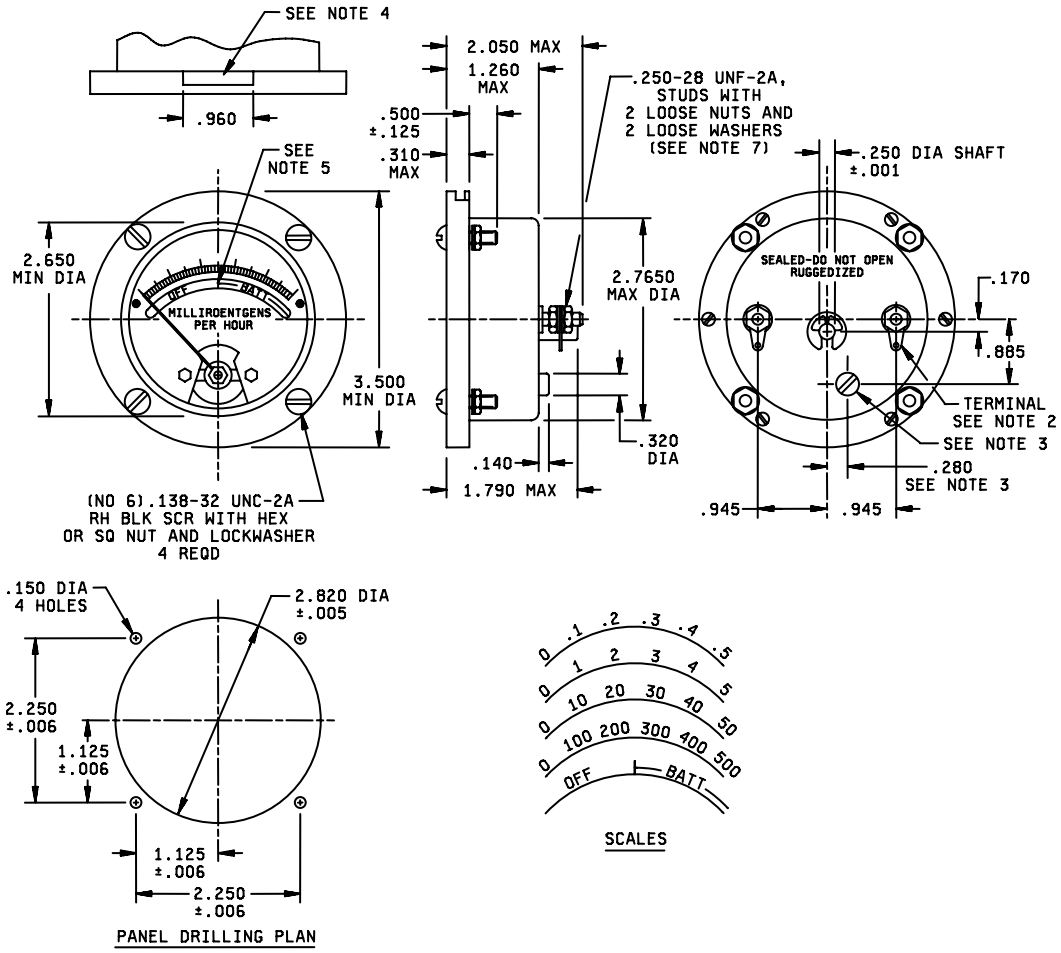
Dielectric withstanding voltage: The meter shall withstand application of 60 Hz at 1,500 volts root mean square (rms), without damage from flashover. This test shall be conducted in accordance with MIL-PRF-10304.

Insulation resistance: Insulation resistance shall be 100 megohms minimum.

Endurance: The scale change shaft shall be rotated for 10,000 cycles (one cycle being defined as that rotation which displays each scale once) after which the water tightness test and shaft rotation and torque tests shall be conducted. This test shall be conducted in accordance with MIL-PRF-10304 before the thermal shock test in group III for qualification inspection and the first test before the thermal shock test in group C qualification verification inspection.

When tested as above: The meter shall operate normally and internal inspection shall show no moisture penetration or loose particles of material. The scale changing shaft shall operate with a torque of no greater than 12 ounce inches.

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Inches	mm	Inches	mm	Inches	mm
.001	.03	.280	7.11	1.260	32.00
.005	.13	.310	7.87	1.790	45.47
.006	.15	.320	8.13	2.050	52.07
.138	3.51	.500	12.70	2.250	57.15
.140	3.56	.885	22.48	2.650	67.31
.150	3.81	.945	24.00	2.760	70.10
.170	4.32	.960	24.38	2.820	71.63
.250	6.35	1.125	28.58	3.500	88.90

FIGURE 1. Multiscale meters, style 36.

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NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerances are ± 0.010 inch for decimals and ± 0.5 (½) degrees for angles.
2. Solder lug terminals shall be used.
3. The zero adjuster shall rotate through 360 degrees and be located 0.280 inch from either side of the vertical center line.
4. This window shall transmit sufficient light to permit easy reading of the scale.
5. When applicable, the scale marking line separating “off” and “battery” shall be located at the mid-scale position.
6. Metric equivalents are given for general information only.
7. Other forms of terminals, as specified in MIL-PRF-10304 (Terminals and connections), are permitted. The specified minimum length applies to stud-type terminals only. Terminals shall be equally spaced, from each side of vertical centerline.

FIGURE 1. Multiscale meters, style 36 - Continued.

TABLE I. Full-scale ranges and colors.

Type designation	MR36M200 SPECR		MR36M201 SPECR		MR36M202 SPECR		MR36M203 SPECR	
Scale units	Counts per minute		Milliroentgens per hour or MREM per hour		Roentgens per hour or REM per hour		Roentgens per hour or REM per hour	
Full scale range (microampere)	20		50		50		500	
Terminal resistance (OHMS)	3,000 ± 20 percent		1,175 ± 20 percent		1,175 ± 20 percent		200 ± 20 percent	
Range selection and Scale colors (FED-STD-595/XXXXX)	Scale		Scale		Scale		Scale	
	Range 1/	Color 2/	Range 1/	Color 2/	Range 1/	Color 2/	Range 1/	Color 2/
	OFF-BATT 0-1M 0-100k 0-10k 0-1k	37875 31158 32246 32356 3/ 33655	OFF-BATT 0-500 0-50 0-5 0-0.5	37875-white 3/ 33655-yellow 37875-white 34552-green 35414-gray	0-10 0-100 0-1K 0-10K Blank	32356 32246 31158 3/ 21105 37875	OFF BATT 0-500 0-50 0-5	37875 37875 3/ 21105 3/ 21158 32356

1/ In the order shown on counterclockwise rotation of the scale change shaft, when viewed from the rear of the meter.
2/ All color codes shown are assigned unique document numbers (see referenced documents on the next page), which are recorded in the format FED-STD-595/XXXXX. The document number search on the ASSIST Online must begin, by using only the base number 595. The actual color chips are then listed on that record and available from the DoDSSP as warehouse items only.
3/ Color shall be lusterless finish.

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The Qualified Products List (QPL) associated with this inactive for new design specification sheet will be maintained until acquisition of the product is no longer required, whereupon the specification and QPL will be canceled.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-10304, this document references the following:

FED-STD-595/21105	FED-STD-595/32246	FED-STD-595/34552
FED-STD-595/21158	FED-STD-595/32356	FED-STD-595/35414
FED-STD-595/31158	FED-STD-595/33655	FED-STD-595/37875

CONCLUDING MATERIAL

Custodians:
Navy – SH
DLA – CC

Preparing activity:
DLA – CC

(Project 6625-0940-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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